

LINUX COMMANDS

CHAPTER 1

SYSTEM INFORMATION COMMANDS

```
shreedhanam2019@cloudshell:~$ uname
Linux
shreedhanam2019@cloudshell:~$ uptime
 18:35:02 up  4:42,  0 user,   load average: 0.27, 0.12, 0.10
shreedhanam2019@cloudshell:~$ hostname
shree
shreedhanam2019@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
shreedhanam2019@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:          16382844      2281464      12506532          1140       1875792       14101380
Swap:              0              0              0
```

```
top - 18:36:25 up  4:43,  0 user,   load average: 0.12, 0.12, 0.09
Tasks:  35 total,   1 running,  34 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  1.8 sy,   0.0 ni, 92.9 id,   5.4 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem : 15998.9 total, 12203.3 free,   2238.0 used,  1831.9 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used. 13760.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	4324	3408	3124	S	0.0	0.0	0:00.08	bash
9	syslog	20	0	222140	3920	2552	S	0.0	0.0	0:00.05	rsyslogd
25	root	20	0	38768	28068	10964	S	0.0	0.2	0:00.84	python
26	root	20	0	6196	2640	2400	S	0.0	0.0	0:00.00	logger
67	root	10	-10	12024	4004	2888	S	0.0	0.0	0:00.03	sshd
372	root	20	0	2169828	85696	61844	S	0.0	0.5	0:00.70	dockerd
441	root	20	0	1234524	12864	8280	S	0.0	0.1	0:00.71	editor-proxy
442	root	20	0	16328	6456	5532	S	0.0	0.0	0:00.01	sudo
447	root	20	0	1226100	2640	1976	S	0.0	0.0	0:00.00	tmux-agent
490	root	20	0	1867212	46748	32832	S	0.0	0.3	0:03.05	containerd
666	shreedh+	10	-10	7960	3620	2140	S	0.0	0.0	0:00.07	tmux
669	shreedh+	10	-10	8928	5820	3932	S	0.0	0.0	0:00.06	bash
721	root	20	0	9196	4060	3684	S	0.0	0.0	0:00.00	runuser
730	shreedh+	20	0	4324	3292	3028	S	0.0	0.0	0:00.00	sh
756	shreedh+	20	0	11.3g	151936	51112	S	0.0	0.9	0:09.31	node
766	root	20	0	2696	1608	1512	S	0.0	0.0	0:00.00	sleep
1643	root	10	-10	13996	9812	8432	S	0.0	0.1	0:00.02	sshd
1645	shreedh+	10	-10	14256	6412	4756	S	0.0	0.0	0:00.00	sshd
1646	shreedh+	10	-10	7340	3720	3472	S	0.0	0.0	0:00.00	bash
1649	shreedh+	10	-10	7604	4392	3760	S	0.0	0.0	0:00.00	bash
1652	shreedh+	20	0	42.4g	744864	62064	S	0.0	4.5	0:33.43	node
1663	shreedh+	20	0	1048516	62028	47804	S	0.0	0.4	0:00.29	node
1686	shreedh+	20	0	1016648	62316	46352	S	0.0	0.4	0:00.32	node
1694	shreedh+	20	0	31.4g	216900	60644	S	0.0	1.3	0:08.53	node
1701	shreedh+	20	0	1853348	616360	49252	S	0.0	3.8	0:09.06	node
1718	shreedh+	20	0	1878572	72852	52012	S	0.0	0.4	0:00.43	cloudcode_cli
1750	shreedh+	10	-10	8928	6000	4108	S	0.0	0.0	0:00.05	bash
2150	shreedh+	10	-10	8928	5984	4096	S	0.0	0.0	0:00.05	bash
2551	root	10	-10	13996	9688	8316	S	0.0	0.1	0:00.01	sshd
2553	shreedh+	10	-10	14256	6464	4808	S	0.0	0.0	0:00.00	sshd
2554	shreedh+	10	-10	7340	3608	3360	S	0.0	0.0	0:00.00	bash
2555	shreedh+	10	-10	7340	3728	3468	S	0.0	0.0	0:00.00	start-shell.sh
2556	shreedh+	10	-10	6880	3336	3040	S	0.0	0.0	0:00.00	tmux
2557	shreedh+	10	-10	8928	5916	4024	S	0.0	0.0	0:00.05	bash

Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

Don't show again Dismiss

```
[[[
1[
2[
3[
Mem[
Swap[

1.3% Tasks: 35, 98 thr, 0 kthr; 1 running
0.7% Load average: 0.03 0.09 0.08
0.7% Uptime: 04:45:16
0.0%
1.92G/15.6G
0K/0K
```

```
Wait [I/O]
PID USER      PRI NI  VIRT  RES  SHR S  CPU%MEM%  TIME+ Command
2351 shreedhana 10 -10 8772 5328 2820 R  1.3  3.0  0:00.16 htop
25 root      20  0 38768 28068 10964 S  0.5  0.2  0:00.86 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --log
2553 shreedhana 10 -10 14256 6464 4808 S  0.5  0.0  0:00.01 sshd: shreedhanam2019@pts/4
10 syslog    20  0 216M 3920 2552 S  0.0  0.0  0:00.02 /usr/sbin/rsyslogd
11 syslog    20  0 216M 3920 2552 S  0.0  0.0  0:00.00 /usr/sbin/rsyslogd
12 syslog    20  0 216M 3920 2552 S  0.0  0.0  0:00.02 /usr/sbin/rsyslogd
25 root      20  0 38768 28068 10964 S  0.0  0.2  0:00.86 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --log
26 root      20  0 6196 2640 2400 S  0.0  0.0  0:00.00 logger -t supervisor
67 root      10 -10 15024 4004 2888 S  0.0  0.0  0:00.03 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
372 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.22 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
413 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.09 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
414 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.08 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
415 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.02 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
416 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
417 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
439 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
441 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.02 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
442 root      20  0 16328 6456 5532 S  0.0  0.0  0:00.01 sudo /google/devshell/tmux-agent
443 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.14 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
444 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.09 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
445 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.03 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
446 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.16 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
447 root      20  0 1197M 2640 1976 S  0.0  0.0  0:00.00 /google/devshell/tmux-agent
448 root      20  0 1197M 2640 1976 S  0.0  0.0  0:00.00 /google/devshell/tmux-agent
449 root      20  0 1197M 2640 1976 S  0.0  0.0  0:00.00 /google/devshell/tmux-agent
450 root      20  0 1197M 2640 1976 S  0.0  0.0  0:00.00 /google/devshell/tmux-agent
451 root      20  0 1197M 2640 1976 S  0.0  0.0  0:00.00 /google/devshell/tmux-agent
462 root      20  0 1205M 12864 8280 S  0.0  0.1  0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/stat
490 root      20  0 1823M 46748 32832 S  0.0  0.3  0:00.05 containerd --config /var/run/docker/containerd/containerd.toml
491 root      20  0 2118M 85696 61844 S  0.0  0.5  0:00.06 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
```

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Info F8Nice F9Kill F10Quit

```
shreedhanam2019@cloudshell:~$ htop
shreedhanam2019@cloudshell:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
shreedhanam2019@cloudshell:~$ arch
x86_64
shreedhanam2019@cloudshell:~$
```

CHAPTER 2

FILE AND DIRECTORY MANAGEMENT

```
shreedhanam2019@cloudshell:~$ ls
README-cloudshell.txt
shreedhanam2019@cloudshell:~$ mkdir demo
shreedhanam2019@cloudshell:~$ touch a.txt
shreedhanam2019@cloudshell:~$ ls
a.txt  demo  README-cloudshell.txt
shreedhanam2019@cloudshell:~$ cd demo
shreedhanam2019@cloudshell:~/demo$ cd ..
shreedhanam2019@cloudshell:~$ mkdir test1
shreedhanam2019@cloudshell:~$ cd test1
shreedhanam2019@cloudshell:~/test1$ pwd
/home/shreedhanam2019/test1
shreedhanam2019@cloudshell:~/test1$ cd ..
shreedhanam2019@cloudshell:~$ pwd
/home/shreedhanam2019
shreedhanam2019@cloudshell:~$ mkdir myfolder
shreedhanam2019@cloudshell:~$ mkdir linuxlab
shreedhanam2019@cloudshell:~$ ls
a.txt  demo  linuxlab  myfolder  README-cloudshell.txt  test1
shreedhanam2019@cloudshell:~$ rmdir linuxlab
shreedhanam2019@cloudshell:~$ mkdir temp
shreedhanam2019@cloudshell:~$ rmdir temp
shreedhanam2019@cloudshell:~$ touch file1.txt
shreedhanam2019@cloudshell:~$ ls
a.txt  demo  file1.txt  myfolder  README-cloudshell.txt  test1
shreedhanam2019@cloudshell:~$ touch notes.txt
shreedhanam2019@cloudshell:~$ ls
a.txt  demo  file1.txt  myfolder  notes.txt  README-cloudshell.txt  test1
shreedhanam2019@cloudshell:~$ rm notes.txt
shreedhanam2019@cloudshell:~$ ls
a.txt  demo  file1.txt  myfolder  README-cloudshell.txt  test1
shreedhanam2019@cloudshell:~$ cp file1.txt file2.txt
```

```
shreedhanam2019@cloudshell:~$ mv a.txt new.txt
shreedhanam2019@cloudshell:~$ ls
copy.txt  file1.txt  myfolder  original.txt  test1
demo      file2.txt  new.txt    README-cloudshell.txt
shreedhanam2019@cloudshell:~$ ln -s original.txt shortcut.txt
shreedhanam2019@cloudshell:~$ ls -l
total 16
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019  0 May 14 18:46 copy.txt
drwxrwxr-x 2 shreedhanam2019 shreedhanam2019 4096 May 14 18:42 demo
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019  0 May 14 18:44 file1.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019  0 May 14 18:45 file2.txt
drwxrwxr-x 2 shreedhanam2019 shreedhanam2019 4096 May 14 18:43 myfolder
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019  0 May 14 18:42 new.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019  0 May 14 18:46 original.txt
-rwxr-xr-x 1 shreedhanam2019 shreedhanam2019  913 May 14 18:41 README-cloudshell.txt
lrwxrwxrwx 1 shreedhanam2019 shreedhanam2019  12 May 14 18:52 shortcut.txt -> original.txt
drwxrwxr-x 2 shreedhanam2019 shreedhanam2019 4096 May 14 18:43 test1
shreedhanam2019@cloudshell:~$ ls -li
131529 copy.txt  131527 file2.txt  131528 original.txt  131524 test1
131522 demo      131525 myfolder   131080 README-cloudshell.txt
131526 file1.txt  131523 new.txt    131530 shortcut.txt
```

CHAPTER 3

FILE VIEWING AND EDITTING

```
shreedhanam2019@cloudshell:~$ touch notes.txt
shreedhanam2019@cloudshell:~$ nano notes.txt
shreedhanam2019@cloudshell:~$ cat notes.txt
Linux is easy
Cloud shell practice
Linux is easy
Learning commands
Google cloud shell
Linux terminal

shreedhanam2019@cloudshell:~$ more notes.txt
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal

shreedhanam2019@cloudshell:~$ less notes.txt
shreedhanam2019@cloudshell:~$ tail notes.txt
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal

shreedhanam2019@cloudshell:~$ tail -2 notes.txt

shreedhanam2019@cloudshell:~$ head notes.txt
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal

shreedhanam2019@cloudshell:~$ ^C
shreedhanam2019@cloudshell:~$ head -3 notes.txt
```

```
shreedhanam2019@cloudshell:~$ head -3 notes.txt
Linux is easy
Cloud shell practice
Learning commands
shreedhanam2019@cloudshell:~$ nano notes.txt
shreedhanam2019@cloudshell:~$ vim notes.txt

[1]+  Stopped                  vim notes.txt
shreedhanam2019@cloudshell:~$ gedit notes.txt
-bash: gedit: command not found
shreedhanam2019@cloudshell:~$ awk '{print}' notes.txt
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal

shreedhanam2019@cloudshell:~$ sed 's/Linux/Ubuntu/' notes.txt
Ubuntu is easy
Cloud shell practice
Learning commands
Google cloud shell
Ubuntu terminal
```

```
GNU nano 7.2
```

```
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal
```

```
Linux is easy
Cloud shell practice
Learning commands
Google cloud shell
Linux terminal
```

```
notes.txt (END)
```

[illegible]

CHAPTER 4

FILE PERMISSIONS AND OWNERSHIP

```
shreedhanam2019@cloudshell:~$ touch sample.txt
shreedhanam2019@cloudshell:~$ ls -l sample.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019 0 May 14 19:14 sample.txt
shreedhanam2019@cloudshell:~$ ls -l sample.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019 0 May 14 19:14 sample.txt
shreedhanam2019@cloudshell:~$ sudo chown $USER sample.txt
sudo: unable to resolve host shree: Name or service not known
shreedhanam2019@cloudshell:~$ ls -l sample.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019 0 May 14 19:14 sample.txt
shreedhanam2019@cloudshell:~$ groups
shreedhanam2019 adm sudo docker
shreedhanam2019@cloudshell:~$ sudo chgrp $(id -gn) sample.txt
sudo: unable to resolve host shree: Name or service not known
shreedhanam2019@cloudshell:~$ ls -l sample.txt
-rw-rw-r-- 1 shreedhanam2019 shreedhanam2019 0 May 14 19:14 sample.txt
shreedhanam2019@cloudshell:~$ umask
0002
shreedhanam2019@cloudshell:~$ stat sample.txt
  File: sample.txt
  Size: 0                Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17    Inode: 131527      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/shreedhanam2019)   Gid: ( 1000/shreedhanam2019)
Access: 2026-05-14 19:14:04.409010662 +0000
Modify: 2026-05-14 19:14:04.409010662 +0000
Change: 2026-05-14 19:15:02.242991049 +0000
 Birth: 2026-05-14 19:14:04.409010662 +0000
shreedhanam2019@cloudshell:~$ getfacl sample.txt
-bash: getfacl: command not found
shreedhanam2019@cloudshell:~$ sudo apt install acl -y
sudo: unable to resolve host shree: Name or service not known
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  acl
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 39.4 kB of archives.
After this operation, 197 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 acl amd64 2.3.2-1build1.1
B]
Fetched 39.4 kB in 0s (114 kB/s)
```

```
Fetched 39.4 kB in 0s (114 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package acl.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../acl_2.3.2-1build1.1_amd64.deb ...
Unpacking acl (2.3.2-1build1.1) ...
Setting up acl (2.3.2-1build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
shreedhanam2019@cloudshell:~$ getfacl sample.txt
# file: sample.txt
# owner: shreedhanam2019
# group: shreedhanam2019
user::rw-
group::rw-
other::r--
```



```

shreedhanam2019@cloudshell:~$ setfacl -m u:$USER:r sample.txt
shreedhanam2019@cloudshell:~$ getfacl sample.txt
# file: sample.txt
# owner: shreedhanam2019
# group: shreedhanam2019
user::rw-
user:shreedhanam2019:r--
group::rw-
mask::rw-
other::r--

shreedhanam2019@cloudshell:~$ lsattr sample.txt
-----e----- sample.txt
shreedhanam2019@cloudshell:~$ sudo chatr +i sample.txt
sudo: unable to resolve host shree: Name or service not known
shreedhanam2019@cloudshell:~$ rm sample.txt
rm: cannot remove 'sample.txt': Operation not permitted
shreedhanam2019@cloudshell:~$ sudo chatr -i sample.txt
sudo: unable to resolve host shree: Name or service not known
shreedhanam2019@cloudshell:~$ rm sample.txt
shreedhanam2019@cloudshell:~$ whoami
shreedhanam2019
shreedhanam2019@cloudshell:~$ id
uid=1000(shreedhanam2019) gid=1000(shreedhanam2019) groups=1000(shreedhanam2019),4(adm)
996(docker)
shreedhanam2019@cloudshell:~$ sudo usermod --help
sudo: unable to resolve host shree: Name or service not known
Usage: usermod [options] LOGIN

Options:
  -a, --append                append the user to the supplemental GROUPS
                              mentioned by the -G option without removing
                              the user from other groups
  -b, --badname               allow bad names
  -c, --comment COMMENT       new value of the GECOS field
  -d, --home HOME_DIR         new home directory for the user account
  -e, --expiredate EXPIRE_DATE set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE     set password inactive after expiration
                              to INACTIVE
  -g, --gid GROUP             force use GROUP as new primary group
  -G, --groups GROUPS         new list of supplementary GROUPS

```

```

shreedhanam2019@cloudshell:~$ sudo usermod --help
sudo: unable to resolve host shree: Name or service not known
Usage: usermod [options] LOGIN

Options:
  -a, --append                append the user to the supplemental GROUPS
                              mentioned by the -G option without removing
                              the user from other groups
  -b, --badname               allow bad names
  -c, --comment COMMENT       new value of the GECOS field
  -d, --home HOME_DIR         new home directory for the user account
  -e, --expiredate EXPIRE_DATE set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE     set password inactive after expiration
                              to INACTIVE
  -g, --gid GROUP             force use GROUP as new primary group
  -G, --groups GROUPS         new list of supplementary GROUPS
  -h, --help                  display this help message and exit
  -l, --login NEW_LOGIN       new value of the login name
  -L, --lock                  lock the user account
  -m, --move-home             move contents of the home directory to the
                              new location (use only with -d)
  -o, --non-unique             allow using duplicate (non-unique) UID
  -p, --password PASSWORD      use encrypted password for the new password
  -P, --prefix PREFIX_DIR      prefix directory where are located the /etc/* files
  -r, --remove                 remove the user from only the supplemental GROUPS
                              mentioned by the -G option without removing
                              the user from other groups
  -R, --root CHROOT_DIR        directory to chroot into
  -s, --shell SHELL           new login shell for the user account
  -u, --uid UID                new UID for the user account
  -U, --unlock                 unlock the user account
  -v, --add-subuids FIRST-LAST add range of subordinate uids
  -V, --del-subuids FIRST-LAST remove range of subordinate uids
  -w, --add-subgids FIRST-LAST add range of subordinate gids
  -W, --del-subgids FIRST-LAST remove range of subordinate gids
  -Z, --selinux-user SEUSER    new SELinux user mapping for the user account

shreedhanam2019@cloudshell:~$ █

```


CHAPTER 5

NETWORKING COMMANDS

```
Cloud Shell
x +
← → ↺ shell.cloud.google.com/?show=ide%2Cterminal
Cloud Shell Editor
cloudshell x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
shreedhanam2019@cloudshell:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 26:f6:5a:51:b7:88 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether e6:04:bf:fb:84:f8 txqueuelen 1000 (Ethernet)
    RX packets 54097 bytes 19146111 (19.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 50146 bytes 54345023 (54.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 37895 bytes 106655955 (106.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37895 bytes 106655955 (106.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

shreedhanam2019@cloudshell:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether e6:04:bf:fb:84:f8 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.4/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 26:f6:5a:51:b7:88 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

```
shreedhanam2019@cloudshell:~$ ping -c 4 google.com
PING google.com (74.125.68.102) 56(84) bytes of data.
64 bytes from sc-in-f102.1e100.net (74.125.68.102): icmp_seq=1 ttl=114 time=1.76 ms
64 bytes from sc-in-f102.1e100.net (74.125.68.102): icmp_seq=2 ttl=114 time=0.754 ms
64 bytes from sc-in-f102.1e100.net (74.125.68.102): icmp_seq=3 ttl=114 time=0.620 ms
64 bytes from sc-in-f102.1e100.net (74.125.68.102): icmp_seq=4 ttl=114 time=0.560 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
rtt min/avg/max/mdev = 0.560/0.924/1.763/0.489 ms
```

```

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
rtt min/avg/max/mdev = 0.560/0.924/1.763/0.489 ms
shreedhanam2019@cloudshell:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:971          localhost:56474         ESTABLISHED
tcp        0      0 localhost:ssh          localhost:44858         ESTABLISHED
tcp        0      0 cs-1094043149067-:51818 sa-in-f138.1e100.:https ESTABLISHED
tcp        0      0 cs-1094043149067-:47298 216.239.32.223:https   TIME_WAIT
tcp        0      0 localhost:57736        localhost:ssh           ESTABLISHED
tcp        0      0 localhost:54404        localhost:970           ESTABLISHED
tcp        0      0 localhost:44858        localhost:ssh           ESTABLISHED
tcp        0      0 localhost:971          localhost:54228         ESTABLISHED
tcp        0      0 localhost:ssh          localhost:57736         ESTABLISHED
tcp        0      0 localhost:56474        localhost:971           ESTABLISHED
tcp        0      0 localhost:ssh          localhost:34166         TIME_WAIT
tcp        0      0 localhost:38140        localhost:970           ESTABLISHED
tcp        0      0 localhost:54228        localhost:971           ESTABLISHED
tcp        0      0 localhost:54384        localhost:ssh           TIME_WAIT
tcp        0      0 localhost:ssh          localhost:33608         TIME_WAIT
tcp6       0      0 cs-1094043149067-de:980 10.111.157.14:47768    ESTABLISHED
tcp6       0      0 cs-1094043149067-de:980 10.111.157.12:41964    ESTABLISHED
tcp6       0      0 cs-1094043149067-de:980 10.111.157.12:44766    ESTABLISHED
tcp6       0      0 cs-1094043149067-de:980 10.111.157.12:44746    ESTABLISHED
tcp6       0      0 localhost:970          localhost:54404         ESTABLISHED
tcp6       0      0 localhost:970          localhost:38140         ESTABLISHED
tcp6       0      0 cs-1094043149067-de:980 10.111.157.12:44756    ESTABLISHED
tcp6       0      0 cs-1094043149067-de:980 10.111.157.12:47884    ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node      Path
unix   3      [ ]        STREAM     CONNECTED    45658
unix   3      [ ]        STREAM     CONNECTED    44808
unix   3      [ ]        STREAM     CONNECTED    82729        /tmp/tmux-1000/default
unix   3      [ ]        STREAM     CONNECTED    45692
unix   3      [ ]        STREAM     CONNECTED    44805
unix   3      [ ]        STREAM     CONNECTED    44804
unix   2      [ ]        DGRAM      CONNECTED    32060
unix   3      [ ]        STREAM     CONNECTED    43673
unix   3      [ ]        STREAM     CONNECTED    45654

```

```

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node      Path
unix   3      [ ]        STREAM     CONNECTED    45658
unix   3      [ ]        STREAM     CONNECTED    44808
unix   3      [ ]        STREAM     CONNECTED    82729        /tmp/tmux-1000/default
unix   3      [ ]        STREAM     CONNECTED    45692
unix   3      [ ]        STREAM     CONNECTED    44805
unix   3      [ ]        STREAM     CONNECTED    44804
unix   2      [ ]        DGRAM      CONNECTED    32060
unix   3      [ ]        STREAM     CONNECTED    43673
unix   3      [ ]        STREAM     CONNECTED    45654
unix   2      [ ]        DGRAM      CONNECTED    31257
unix   3      [ ]        STREAM     CONNECTED    45660
unix   2      [ ]        STREAM     CONNECTED    86134
unix   3      [ ]        STREAM     CONNECTED    45656
unix   3      [ ]        STREAM     CONNECTED    44803
unix   3      [ ]        STREAM     CONNECTED    43680
unix   3      [ ]        STREAM     CONNECTED    44569
unix   3      [ ]        STREAM     CONNECTED    43678
unix   3      [ ]        STREAM     CONNECTED    45690
unix   3      [ ]        STREAM     CONNECTED    44807
unix   3      [ ]        STREAM     CONNECTED    43676
unix   3      [ ]        STREAM     CONNECTED    82353
unix   3      [ ]        STREAM     CONNECTED    84743
unix   3      [ ]        STREAM     CONNECTED    45696
unix   3      [ ]        STREAM     CONNECTED    45695
unix   3      [ ]        STREAM     CONNECTED    43677
unix   3      [ ]        STREAM     CONNECTED    43675
unix   3      [ ]        STREAM     CONNECTED    45694
unix   3      [ ]        STREAM     CONNECTED    44806
unix   2      [ ]        DGRAM      CONNECTED    30554
unix   2      [ ]        DGRAM      CONNECTED    44542
unix   3      [ ]        STREAM     CONNECTED    82354
unix   3      [ ]        STREAM     CONNECTED    82350
unix   3      [ ]        STREAM     CONNECTED    43679
unix   3      [ ]        STREAM     CONNECTED    30661
unix   3      [ ]        STREAM     CONNECTED    85737
unix   3      [ ]        STREAM     CONNECTED    45653
unix   3      [ ]        STREAM     CONNECTED    45691
unix   3      [ ]        STREAM     CONNECTED    84742
unix   3      [ ]        STREAM     CONNECTED    44548

```

```
unix 9      [ ]      DGRAM    CONNECTED 31064    /dev/log
unix 2      [ ]      DGRAM    CONNECTED 84737
unix 3      [ ]      STREAM   CONNECTED 30665
unix 3      [ ]      STREAM   CONNECTED 45655
unix 3      [ ]      STREAM   CONNECTED 45697
unix 3      [ ]      STREAM   CONNECTED 32141    /var/run/docker/containerd/containerd.sock
unix 3      [ ]      STREAM   CONNECTED 45693
unix 3      [ ]      STREAM   CONNECTED 45657
unix 3      [ ]      STREAM   CONNECTED 44547
unix 2      [ ]      DGRAM    CONNECTED 32338
unix 3      [ ]      STREAM   CONNECTED 82349
unix 2      [ ]      DGRAM    CONNECTED 29466
unix 3      [ ]      STREAM   CONNECTED 45659
unix 3      [ ]      STREAM   CONNECTED 32147    /var/run/docker/containerd/containerd.sock
unix 3      [ ]      STREAM   CONNECTED 82351
unix 3      [ ]      STREAM   CONNECTED 43674
unix 3      [ ]      STREAM   CONNECTED 44570
unix 2      [ ]      STREAM   CONNECTED 45588
```

```
shreedhanam2019@cloudshell:~$ ss
Netid      State      Rcvd-Q     Send-Q      Local Address:Port      Peer Address:Port
u_str      ESTAB      0           0            * 45658                  * 45657
u_str      ESTAB      0           0            * 44808                  * 44807
u_str      ESTAB      0           0            /tmp/tmux-1000/default 82729      * 85737
u_str      ESTAB      0           0            * 45692                  * 45693
u_str      ESTAB      0           0            * 44805                  * 44806
u_str      ESTAB      0           0            * 44804                  * 44803
u_dgr      ESTAB      0           0            * 32060                  * 31064
u_str      ESTAB      0           0            * 43673                  * 43674
u_str      ESTAB      0           0            * 45654                  * 45653
u_dgr      ESTAB      0           0            * 31257                  * 31064
```

```
shreedhanam2019@cloudshell:~$ sudo apt install traceroute -y
sudo: unable to resolve host shree: Name or service not known
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (181 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121200 files and directories currently installed.)
Preparing to unpack .../traceroute_1:2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.db.1.gz (of link group traceroute) doesn't exist
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.db.1.gz (of link group traceroute6) doesn't exist
update-alternatives: using /usr/bin/ift.db to provide /usr/bin/ift (ift) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/ift.db.1.gz because associated file /usr/share/man/man1/ift.db.1.gz (of link group ift) doesn't exist
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.db.1.gz (of link group traceproto) doesn't exist
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man8/tcptraceroute.8.gz because associated file /usr/share/man/man8/tcptraceroute.db.8.gz (of link group tcptraceroute) doesn't exist
Processing triggers for man-db (2.12.0-4build2) ...
shreedhanam2019@cloudshell:~$ traceroute google.com
traceroute to google.com (64.233.170.139), 30 hops max, 60 byte packets
 1 10.88.0.1 (10.88.0.1) 0.142 ms 0.017 ms 0.011 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
```

```
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * sg-in-f139.1e100.net (64.233.170.139) 1.150 ms
shreedhanam2019@cloudshell:~$ wget https://example.com
--2026-05-14 19:26:49-- https://example.com/
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10::ac42:93f3, ...
Connecting to example.com (example.com)[104.20.23.154]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html      [ <=>          ]    528  --.-KB/s    in 0s

2026-05-14 19:26:49 (69.5 MB/s) - 'index.html' saved [528]

shreedhanam2019@cloudshell:~$ ls
index.html  notes.txt  README-cloudshell.txt
shreedhanam2019@cloudshell:~$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-uGechBfiXW4A7Cwlnz7I-g'
http://report-uri https://csp.withgoogle.com/csp/gws/other-hp
date: Thu, 14 May 2026 19:27:04 GMT
expires: Sat, 13 Jun 2026 19:27:04 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
```

```
shreedhanam2019@cloudshell:~$ sudo apt install dnsutils -y
sudo: unable to resolve host shree: Name or service not known
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
dnsutils is already the newest version (1:9.18.39-0ubuntu0.24.04.3).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
shreedhanam2019@cloudshell:~$ nslookup google.com
Server:      169.254.169.254
Address:     169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 64.233.170.139
Name:   google.com
Address: 64.233.170.101
Name:   google.com
Address: 64.233.170.113
Name:   google.com
Address: 64.233.170.102
Name:   google.com
Address: 64.233.170.100
Name:   google.com
Address: 64.233.170.138
Name:   google.com
Address: 2404:6800:4003:c11::64
Name:   google.com
Address: 2404:6800:4003:c11::71
Name:   google.com
Address: 2404:6800:4003:c11::8a
Name:   google.com
Address: 2404:6800:4003:c11::66
```

```
shreedhanam2019@cloudshell:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6557
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1
```

```
shreedhanam2019@cloudshell:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6557
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

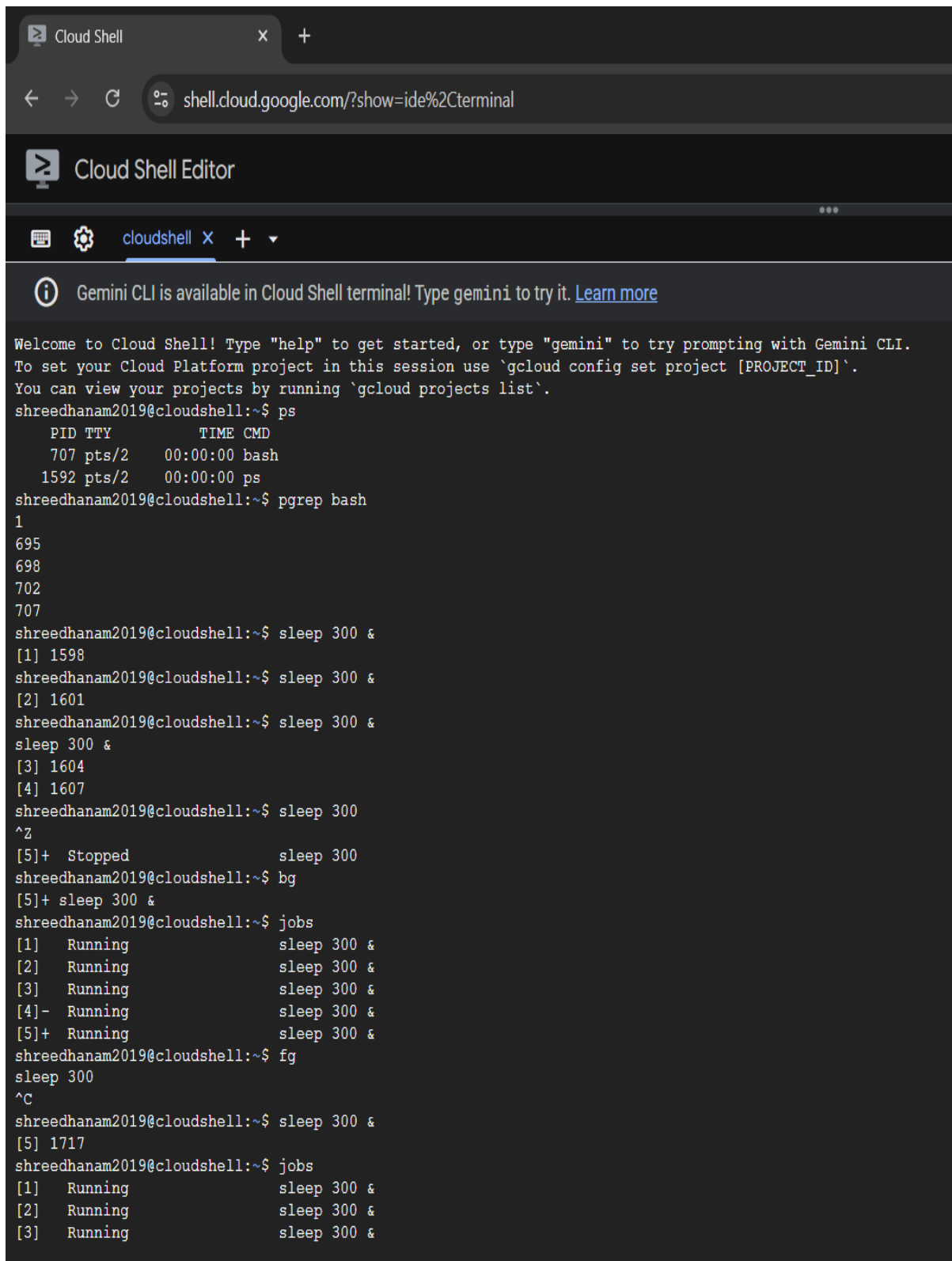
;; ANSWER SECTION:
google.com.                 294     IN      A      64.233.170.139
google.com.                 294     IN      A      64.233.170.101
google.com.                 294     IN      A      64.233.170.113
google.com.                 294     IN      A      64.233.170.102
google.com.                 294     IN      A      64.233.170.100
google.com.                 294     IN      A      64.233.170.138

;; Query time: 2 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 19:27:24 UTC 2026
;; MSG SIZE rcvd: 135
```

```
shreedhanam2019@cloudshell:~$ dig google.com +short
64.233.170.139
64.233.170.101
64.233.170.113
64.233.170.102
64.233.170.100
64.233.170.138
shreedhanam2019@cloudshell:~$
```

CHAPTER 6

PROCESS MANAGEMENT



The screenshot shows a Cloud Shell terminal window with the following content:

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

cloudshell X + v

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
shreedhanam2019@cloudshell:~$ ps
  PID TTY          TIME CMD
   707 pts/2    00:00:00 bash
  1592 pts/2    00:00:00 ps
shreedhanam2019@cloudshell:~$ pgrep bash
1
695
698
702
707
shreedhanam2019@cloudshell:~$ sleep 300 &
[1] 1598
shreedhanam2019@cloudshell:~$ sleep 300 &
[2] 1601
shreedhanam2019@cloudshell:~$ sleep 300 &
sleep 300 &
[3] 1604
[4] 1607
shreedhanam2019@cloudshell:~$ sleep 300
^Z
[5]+  Stopped                  sleep 300
shreedhanam2019@cloudshell:~$ bg
[5]+ sleep 300 &
shreedhanam2019@cloudshell:~$ jobs
[1]  Running                  sleep 300 &
[2]  Running                  sleep 300 &
[3]  Running                  sleep 300 &
[4]- Running                  sleep 300 &
[5]+ Running                  sleep 300 &
shreedhanam2019@cloudshell:~$ fg
sleep 300
^C
shreedhanam2019@cloudshell:~$ sleep 300 &
[5] 1717
shreedhanam2019@cloudshell:~$ jobs
[1]  Running                  sleep 300 &
[2]  Running                  sleep 300 &
[3]  Running                  sleep 300 &
```

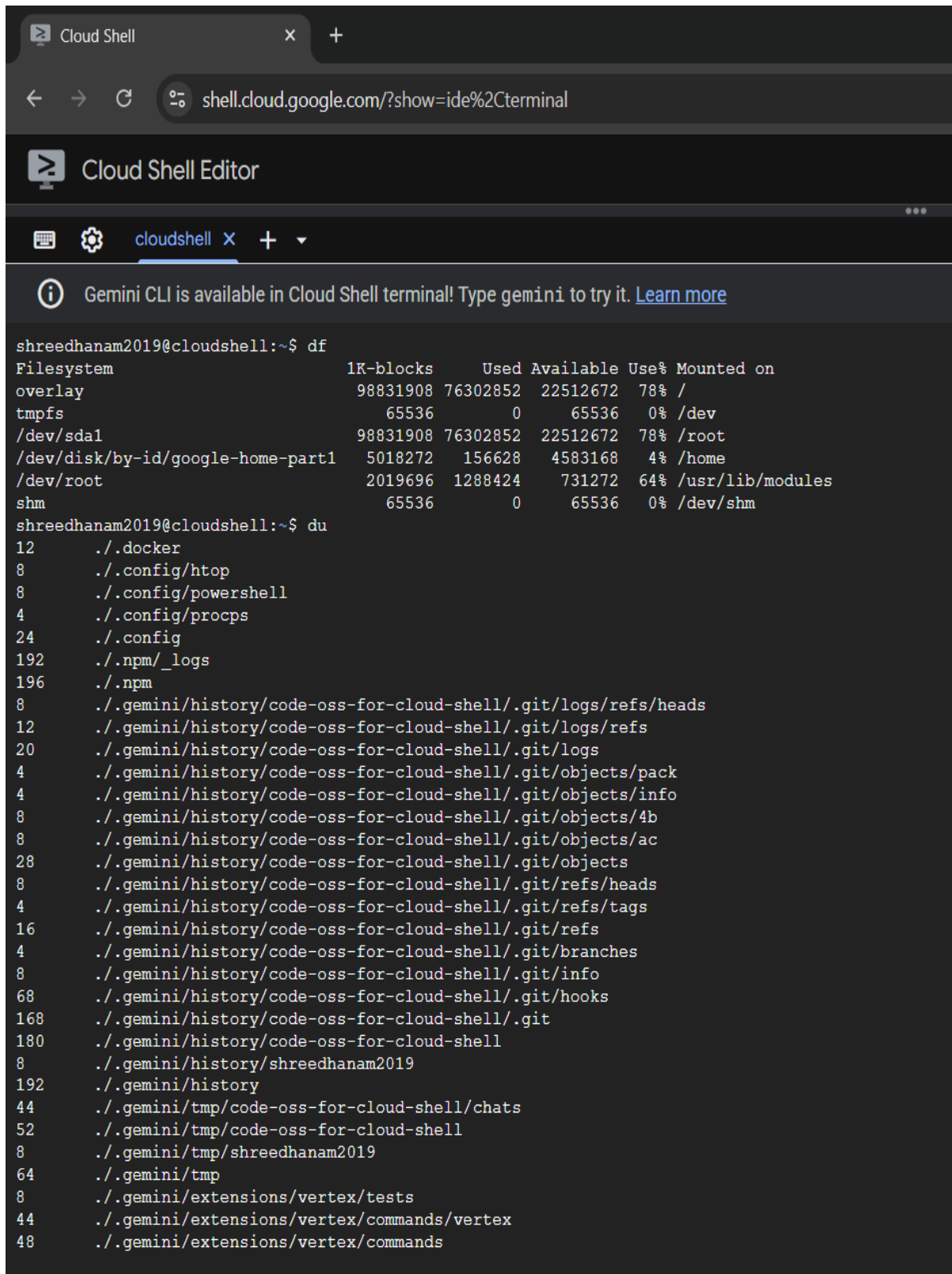
```

shreedhanam2019@cloudshell:~$ jobs
[1]   Running                  sleep 300 &
[2]   Running                  sleep 300 &
[3]   Running                  sleep 300 &
[4]-  Running                  sleep 300 &
[5]+  Running                  sleep 300 &
shreedhanam2019@cloudshell:~$ nice -n 10 sleep 300 &
[6] 1722
shreedhanam2019@cloudshell:~$ ps -l
F S  UID      PID      PPID  C  PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S   1000      707      706  0   70  -10 - 2232 do_wai pts/2    00:00:00 bash
0 S   1000     1598      707  0   70  -10 - 1422 hrtime pts/2    00:00:00 sleep
0 S   1000     1601      707  0   70  -10 - 1422 hrtime pts/2    00:00:00 sleep
0 S   1000     1604      707  0   70  -10 - 1422 hrtime pts/2    00:00:00 sleep
0 S   1000     1607      707  0   70  -10 - 1422 hrtime pts/2    00:00:00 sleep
0 S   1000     1717      707  0   70  -10 - 1422 hrtime pts/2    00:00:00 sleep
0 S   1000     1722      707  0   80   0 - 1422 hrtime pts/2    00:00:00 sleep
0 R   1000     1727      707  0   70  -10 - 2730 -      pts/2    00:00:00 ps
shreedhanam2019@cloudshell:~$ ps -ef | grep sleep
root      1          0  0 00:39 ?        00:00:00 /bin/bash /google/scripts/onrun.sh sleep infinity
root      690        1  0 00:39 ?        00:00:00 sleep infinity
shreedh+  1598      707  0 00:42 pts/2    00:00:00 sleep 300
shreedh+  1601      707  0 00:42 pts/2    00:00:00 sleep 300
shreedh+  1604      707  0 00:42 pts/2    00:00:00 sleep 300
shreedh+  1607      707  0 00:42 pts/2    00:00:00 sleep 300
shreedh+  1717      707  0 00:44 pts/2    00:00:00 sleep 300
shreedh+  1722      707  0 00:44 pts/2    00:00:00 sleep 300
shreedh+  1731      707  0 00:45 pts/2    00:00:00 grep --color=auto sleep
shreedhanam2019@cloudshell:~$ renice 5 -p PID
renice: bad process ID value: PID
shreedhanam2019@cloudshell:~$

```

CHAPTER 7

DISK MANAGEMENT



The screenshot shows a Cloud Shell terminal window. The browser address bar displays `shell.cloud.google.com/?show=ide%2Cterminal`. The terminal interface includes a title bar "Cloud Shell Editor" and a toolbar with icons for keyboard shortcuts, settings, and tabs. A notification banner at the top states: "Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)".

The terminal session shows the following commands and output:

```
shreedhanam2019@cloudshell:~$ df
Filesystem                1K-blocks      Used Available Use% Mounted on
overlay                   98831908  76302852   22512672   78% /
tmpfs                     65536           0     65536    0% /dev
/dev/sda1                 98831908  76302852   22512672   78% /root
/dev/disk/by-id/google-home-part1  5018272    156628    4583168    4% /home
/dev/root                 2019696   1288424     731272   64% /usr/lib/modules
shm                       65536           0     65536    0% /dev/shm

shreedhanam2019@cloudshell:~$ du
12  ./docker
8   ./config/htop
8   ./config/powershell
4   ./config/procps
24  ./config
192 ./npm/_logs
196 ./npm
8   ./gemini/history/code-oss-for-cloud-shell/.git/logs/refs/heads
12  ./gemini/history/code-oss-for-cloud-shell/.git/logs/refs
20  ./gemini/history/code-oss-for-cloud-shell/.git/logs
4   ./gemini/history/code-oss-for-cloud-shell/.git/objects/pack
4   ./gemini/history/code-oss-for-cloud-shell/.git/objects/info
8   ./gemini/history/code-oss-for-cloud-shell/.git/objects/4b
8   ./gemini/history/code-oss-for-cloud-shell/.git/objects/ac
28  ./gemini/history/code-oss-for-cloud-shell/.git/objects
8   ./gemini/history/code-oss-for-cloud-shell/.git/refs/heads
4   ./gemini/history/code-oss-for-cloud-shell/.git/refs/tags
16  ./gemini/history/code-oss-for-cloud-shell/.git/refs
4   ./gemini/history/code-oss-for-cloud-shell/.git/branches
8   ./gemini/history/code-oss-for-cloud-shell/.git/info
68  ./gemini/history/code-oss-for-cloud-shell/.git/hooks
168 ./gemini/history/code-oss-for-cloud-shell/.git
180 ./gemini/history/code-oss-for-cloud-shell
8   ./gemini/history/shreedhanam2019
192 ./gemini/history
44  ./gemini/tmp/code-oss-for-cloud-shell/chats
52  ./gemini/tmp/code-oss-for-cloud-shell
8   ./gemini/tmp/shreedhanam2019
64  ./gemini/tmp
8   ./gemini/extensions/vertex/tests
44  ./gemini/extensions/vertex/commands/vertex
48  ./gemini/extensions/vertex/commands
```




Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

```
48  ./gemini/extensions/vertex/commands
44  ./gemini/extensions/vertex/src/vertex/prompt_optimizer/docs
100 ./gemini/extensions/vertex/src/vertex/prompt_optimizer
120 ./gemini/extensions/vertex/src/vertex
124 ./gemini/extensions/vertex/src
8   ./gemini/extensions/vertex/.git/logs/refs/heads
8   ./gemini/extensions/vertex/.git/logs/refs/remotes/origin
12  ./gemini/extensions/vertex/.git/logs/refs/remotes
24  ./gemini/extensions/vertex/.git/logs/refs
32  ./gemini/extensions/vertex/.git/logs
224 ./gemini/extensions/vertex/.git/objects/pack
4   ./gemini/extensions/vertex/.git/objects/info
232 ./gemini/extensions/vertex/.git/objects
8   ./gemini/extensions/vertex/.git/refs/heads
8   ./gemini/extensions/vertex/.git/refs/remotes/origin
12  ./gemini/extensions/vertex/.git/refs/remotes
4   ./gemini/extensions/vertex/.git/refs/tags
28  ./gemini/extensions/vertex/.git/refs
4   ./gemini/extensions/vertex/.git/branches
8   ./gemini/extensions/vertex/.git/info
68  ./gemini/extensions/vertex/.git/hooks
404 ./gemini/extensions/vertex/.git
8   ./gemini/extensions/vertex/.github/workflows
12  ./gemini/extensions/vertex/.github
1208 ./gemini/extensions/vertex
1216 ./gemini/extensions
1496 ./gemini
8   ./vscode
4   ./local/share/nano
8   ./local/share
12  ./local
8   ./cache/Microsoft/DeveloperTools
12  ./cache/Microsoft
128484 ./cache/cloud-code/cloudcode_cli/cloudcode_cli/fba419f4
128488 ./cache/cloud-code/cloudcode_cli/cloudcode_cli
128492 ./cache/cloud-code/cloudcode_cli
19760 ./cache/cloud-code/api-metadata
148260 ./cache/cloud-code
148276 ./cache
8   ./codeoss/data/Machine
4   ./codeoss/data/User/History
```



Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

```
4      ../codeoss/data/User/History
4      ../codeoss/data/User/globalStorage/vscode.json-language-features/json-schema-cache
8      ../codeoss/data/User/globalStorage/vscode.json-language-features
12     ../codeoss/data/User/globalStorage
20     ../codeoss/data/User
8      ../codeoss/data/logs/20260514T150904/exthost4/vscode.git
8      ../codeoss/data/logs/20260514T150904/exthost4/vscode.github-authentication
4      ../codeoss/data/logs/20260514T150904/exthost4/vscode.json-language-features
336    ../codeoss/data/logs/20260514T150904/exthost4/output_logging_20260514T152802
8      ../codeoss/data/logs/20260514T150904/exthost4/vscode.github
372    ../codeoss/data/logs/20260514T150904/exthost4
8      ../codeoss/data/logs/20260514T150904/exthost5/vscode.git
8      ../codeoss/data/logs/20260514T150904/exthost5/vscode.github-authentication
444    ../codeoss/data/logs/20260514T150904/exthost5/output_logging_20260514T155240
4      ../codeoss/data/logs/20260514T150904/exthost5/vscode.json-language-features
8      ../codeoss/data/logs/20260514T150904/exthost5/vscode.github
480    ../codeoss/data/logs/20260514T150904/exthost5
8      ../codeoss/data/logs/20260514T150904/exthost1
8      ../codeoss/data/logs/20260514T150904/exthost2/vscode.git
8      ../codeoss/data/logs/20260514T150904/exthost2/vscode.github-authentication
4      ../codeoss/data/logs/20260514T150904/exthost2/vscode.json-language-features
8      ../codeoss/data/logs/20260514T150904/exthost2/vscode.github
244    ../codeoss/data/logs/20260514T150904/exthost2/output_logging_20260514T151524
280    ../codeoss/data/logs/20260514T150904/exthost2
8      ../codeoss/data/logs/20260514T150904/exthost3/vscode.git
8      ../codeoss/data/logs/20260514T150904/exthost3/vscode.github-authentication
4      ../codeoss/data/logs/20260514T150904/exthost3/vscode.json-language-features
8      ../codeoss/data/logs/20260514T150904/exthost3/vscode.github
152    ../codeoss/data/logs/20260514T150904/exthost3/output_logging_20260514T152122
188    ../codeoss/data/logs/20260514T150904/exthost3
1340   ../codeoss/data/logs/20260514T150904
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.git
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.github-authentication
4      ../codeoss/data/logs/20260514T180159/exthost1/vscode.json-language-features
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.github
8      ../codeoss/data/logs/20260514T180159/exthost1/output_logging_20260514T180206
52     ../codeoss/data/logs/20260514T180159/exthost1
4      ../codeoss/data/logs/20260514T180159/exthost2/vscode.git
8      ../codeoss/data/logs/20260514T180159/exthost2/vscode.github-authentication
4      ../codeoss/data/logs/20260514T180159/exthost2/vscode.github
24     ../codeoss/data/logs/20260514T180159/exthost2
```



Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

```
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.git
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.github-authentication
4      ../codeoss/data/logs/20260514T180159/exthost1/vscode.json-language-features
8      ../codeoss/data/logs/20260514T180159/exthost1/vscode.github
8      ../codeoss/data/logs/20260514T180159/exthost1/output_logging_20260514T180206
52     ../codeoss/data/logs/20260514T180159/exthost1
4      ../codeoss/data/logs/20260514T180159/exthost2/vscode.git
8      ../codeoss/data/logs/20260514T180159/exthost2/vscode.github-authentication
4      ../codeoss/data/logs/20260514T180159/exthost2/vscode.github
24     ../codeoss/data/logs/20260514T180159/exthost2
648    ../codeoss/data/logs/20260514T180159/exthost3/output_logging_20260514T182103
8      ../codeoss/data/logs/20260514T180159/exthost3/vscode.git
8      ../codeoss/data/logs/20260514T180159/exthost3/vscode.github-authentication
4      ../codeoss/data/logs/20260514T180159/exthost3/vscode.json-language-features
8      ../codeoss/data/logs/20260514T180159/exthost3/vscode.github
684    ../codeoss/data/logs/20260514T180159/exthost3
768    ../codeoss/data/logs/20260514T180159
448    ../codeoss/data/logs/20260514T101315/exthost1/output_logging_20260514T101325
8      ../codeoss/data/logs/20260514T101315/exthost1/vscode.git
8      ../codeoss/data/logs/20260514T101315/exthost1/vscode.github-authentication
4      ../codeoss/data/logs/20260514T101315/exthost1/vscode.json-language-features
8      ../codeoss/data/logs/20260514T101315/exthost1/vscode.github
484    ../codeoss/data/logs/20260514T101315/exthost1
492    ../codeoss/data/logs/20260514T101315
248    ../codeoss/data/logs/20260515T003944/exthost1/output_logging_20260515T003951
8      ../codeoss/data/logs/20260515T003944/exthost1/vscode.git
8      ../codeoss/data/logs/20260515T003944/exthost1/vscode.github-authentication
4      ../codeoss/data/logs/20260515T003944/exthost1/vscode.json-language-features
8      ../codeoss/data/logs/20260515T003944/exthost1/vscode.github
284    ../codeoss/data/logs/20260515T003944/exthost1
248    ../codeoss/data/logs/20260515T003944/exthost2/output_logging_20260515T004349
8      ../codeoss/data/logs/20260515T003944/exthost2/vscode.git
8      ../codeoss/data/logs/20260515T003944/exthost2/vscode.github-authentication
4      ../codeoss/data/logs/20260515T003944/exthost2/vscode.json-language-features
8      ../codeoss/data/logs/20260515T003944/exthost2/vscode.github
284    ../codeoss/data/logs/20260515T003944/exthost2
576    ../codeoss/data/logs/20260515T003944
8      ../codeoss/data/logs/20260514T054845/exthost1/vscode.git
8      ../codeoss/data/logs/20260514T054845/exthost1/vscode.github-authentication
4      ../codeoss/data/logs/20260514T054845/exthost1/vscode.json-language-features
1216   ../codeoss/data/logs/20260514T054845/exthost1/output_logging_20260514T054852
```

```
Cloud Shell
x +
Ask Gemini
Paused
Cloud Shell Editor
cloudshell x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more
Don't show again Dismiss

8 ./codeoss/data/logs/20260514T054845/axhost1/vscode.github
1252 ./codeoss/data/logs/20260514T054845/axhost1
1260 ./codeoss/data/logs/20260514T054845
436 ./codeoss/data/logs/20260514T045453/axhost1/output_logging_20260514T045515
8 ./codeoss/data/logs/20260514T045453/axhost1/vscode.git
8 ./codeoss/data/logs/20260514T045453/axhost1/vscode.github-authentication
4 ./codeoss/data/logs/20260514T045453/axhost1/vscode.json-language-features
8 ./codeoss/data/logs/20260514T045453/axhost1/vscode.github
472 ./codeoss/data/logs/20260514T045453/axhost1
480 ./codeoss/data/logs/20260514T045453
4920 ./codeoss/data/logs
1552 ./codeoss/data/CachedProfilesData/_default_profile_
1556 ./codeoss/data/CachedProfilesData
6512 ./codeoss/data
8 ./codeoss/extensions
6524 ./codeoss
156604

shreedhanam2019@cloudshell:~$ sudo fdisk -l
sudo: fdisk: command not found
shreedhanam2019@cloudshell:~$ mkfs --help

Usage:
mkfs [options] [-t <type>] [-f] [-o options] <device> [<size>]

Make a Linux filesystem.

Options:
-t, --type=<type> filesystem type; when unspecified, ext2 is used
-fs-options parameters for the real filesystem builder
<device> path to the device to be used
<size> number of blocks to be used on the device
-V, --verbose explain what is being done;
-V, --specifying -V more than once will cause a dry-run
-h, --help display this help
-V, --version display version

For more details see mkfs(8).
shreedhanam2019@cloudshell:~$ mount
overlay on / type overlay (rw,relatime,lowerdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshots/318/work)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
/dev/sdal on /root type ext4 (rw,relatime,commit=30)
/dev/disk/by-id/google-home-part1 on /home type ext4 (rw,noatime,journal_checksum,errors=remount-ro,data=ordered)
/dev/dm-0 on /usr/lib/modules type ext2 (ro,relatime)
/dev/sdal on /etc/hosts type ext4 (rw,relatime,commit=30)
/dev/sdal on /etc/hostname type ext4 (rw,relatime,commit=30)
/dev/sdal on /etc/resolv.conf type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sdal on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sdal on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sdal on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sdal on /run/google/devshell type ext4 (rw,relatime,commit=30)
none on /sys/kernel/security type securityfs (rw,relatime)
shreedhanam2019@cloudshell:~$ mount --help

Usage:
mount [-hW]
mount -a [options]
mount [options] <source> | <directory>

Unmount filesystems.

Options:
-a, --all unmount all filesystems
-A, --all-targets unmount all mountpoints for the given device in the
current namespace
-c, --no-canonicalize don't canonicalize paths
-d, --detach-loop if mounted loop device, also free this loop device
--fake dry run: skip the mount(2) syscall
-f, --force force unmount (in case of an unreachable NFS system)
-i, --internal-only don't call the mount.<type> helpers
-n, --no-mtab don't write to /etc/mtab
-l, --lazy detach the filesystem now, clean up things later
```

```
Cloud Shell
x +
Ask Gemini
Paused
Cloud Shell Editor
cloudshell x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more
Don't show again Dismiss

fs:126/fs:125/fs:123/fs:119/fs:118/fs:116/fs:114/fs:110/fs:107/fs:106/fs:105/fs:104/fs:102/fs:99/fs:97/fs:95/fs:94/fs:91/fs:90/fs:85/fs:84/fs:83/fs:82/fs:80/fs:78/fs:69/fs:68/fs:67/f
s:66/fs:65/fs:64/fs:63/fs:62/fs:6/fs:6/fs:upperdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshots/318/work,workdir=/var/lib/containerd/io.containerd.snapshotter
.v1.gcfs/snapshots/318/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
/dev/sdal on /root type ext4 (rw,relatime,commit=30)
/dev/disk/by-id/google-home-part1 on /home type ext4 (rw,noatime,journal_checksum,errors=remount-ro,data=ordered)
/dev/dm-0 on /usr/lib/modules type ext2 (ro,relatime)
/dev/sdal on /etc/hosts type ext4 (rw,relatime,commit=30)
/dev/sdal on /etc/hostname type ext4 (rw,relatime,commit=30)
/dev/sdal on /etc/resolv.conf type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sdal on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sdal on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sdal on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sdal on /run/google/devshell type ext4 (rw,relatime,commit=30)
none on /sys/kernel/security type securityfs (rw,relatime)
shreedhanam2019@cloudshell:~$ mount --help

Usage:
mount [-hW]
mount -a [options]
mount [options] <source> | <directory>

Unmount filesystems.

Options:
-a, --all unmount all filesystems
-A, --all-targets unmount all mountpoints for the given device in the
current namespace
-c, --no-canonicalize don't canonicalize paths
-d, --detach-loop if mounted loop device, also free this loop device
--fake dry run: skip the mount(2) syscall
-f, --force force unmount (in case of an unreachable NFS system)
-i, --internal-only don't call the mount.<type> helpers
-n, --no-mtab don't write to /etc/mtab
-l, --lazy detach the filesystem now, clean up things later
```


CHAPTER 8

SYSTEM MONITORING AND PERFORMANCE

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

top - 00:59:02 up 2:19, 0 user, load average: 0.03, 0.03, 0.05
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.3 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.1 st
MiB Mem : 15998.9 total, 12175.1 free, 2269.0 used, 1829.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 13729.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM    TIME+  COMMAND
  140 root        20   0 1234720  1284    822   S  0.3   0.1   0:00.62 editor-proxy
2549 shreedh+  10 -10  11928   5900  3684   R  0.3   0.0   0:00.03 top
    1 root        20   0   4324   3444  3168   S  0.0   0.0   0:00.07 bash
    9 syslog     20   0 222140   3816  2456   S  0.0   0.0   0:00.03 rsyslogd
   25 root        20   0   38768  28072 10972   S  0.0   0.2   0:00.64 python
   26 root        20   0   6196   2668  2432   S  0.0   0.0   0:00.00 logger
   67 root        10 -10  12024   4200  3088   S  0.0   0.0   0:00.02 sshd
  371 root        20   0 2095840  85140 61208   S  0.0   0.5   0:00.53 dockerd
  441 root        20   0   16328   6432  5804   S  0.0   0.0   0:00.02 sudo
  446 root        20   0 1226356  2764   1976   S  0.0   0.0   0:00.00 tmux-agent
  490 root        20   0 1867212 45564 31544   S  0.0   0.3   0:01.97 containerd
  690 root        20   0   2696   1612  1512   S  0.0   0.0   0:00.00 sleep
  706 shreedh+  10 -10   7692  3236  2116   S  0.0   0.0   0:00.07 tmux
  707 shreedh+  10 -10   8928  5936  4028   S  0.0   0.0   0:00.08 bash
  753 root        20   0   9196   4012  3636   S  0.0   0.0   0:00.00 runuser
  754 shreedh+  20   0   4324   3236  2972   S  0.0   0.0   0:00.00 sh
  775 shreedh+  20   0   11.2g 145132 51076   S  0.0   0.9   0:05.66 node
1617 root        10 -10  13996   9704  8332   S  0.0   0.1   0:00.01 sshd
1621 shreedh+  10 -10  14256   6336  4684   S  0.0   0.0   0:00.00 sshd
1623 shreedh+  10 -10   7340   3760  3512   S  0.0   0.0   0:00.00 bash
1626 shreedh+ -10 -10   7604   4432  3792   S  0.0   0.0   0:00.00 bash
1632 shreedh+  20   0 42.5g 801364 62228   S  0.0   4.9   0:31.46 node
1643 shreedh+  20   0 1048516 61652 47880   S  0.0   0.4   0:00.33 node
1665 shreedh+  20   0 1016648 61328 46160   S  0.0   0.4   0:00.31 node
1673 shreedh+  20   0 31.4g 216892 60696   S  0.0   1.3   0:07.64 node
1687 shreedh+  20   0 1853348 616192 49300   S  0.0   3.8   0:08.08 node
1694 shreedh+  20   0 1877804 71956 51436   S  0.0   0.4   0:00.38 cloudcode_cli
1745 shreedh+  10 -10   8928  5964  4064   S  0.0   0.0   0:00.07 bash
2173 root        10 -10  13996   9748  8372   S  0.0   0.1   0:00.01 sshd
2175 shreedh+  10 -10  14256   6416  4764   S  0.0   0.0   0:00.00 sshd
2176 shreedh+  10 -10   7340   3780  3532   S  0.0   0.0   0:00.00 bash
2177 shreedh+  10 -10   7340   3728  3468   S  0.0   0.0   0:00.00 start-shell.sh
2178 shreedh+  10 -10   6880   3432  3132   S  0.0   0.0   0:00.00 tmux
2179 shreedh+  10 -10   8928  5868  3976   S  0.0   0.0   0:00.04 bash
```

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

shreedhanam2019@cloudshell:~$ sudo apt install htop -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
htop is already the newest version (3.3.0-4build1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
shreedhanam2019@cloudshell:~$ htop
shreedhanam2019@cloudshell:~$ sudo apt install sysstat -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  lssap
The following NEW packages will be installed:
  sysstat
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 489 kB of archives.
After this operation, 1,487 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 sysstat amd64 12.6.1-2 [489 kB]
Fetched 489 kB in 0s (9,650 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package sysstat.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../sysstat_12.6.1-2_amd64.deb ...
Unpacking sysstat (12.6.1-2) ...
Setting up sysstat (12.6.1-2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline

Creating config file /etc/default/sysstat with new version
update-alternatives: using /usr/bin/sar.sysstat to provide /usr/bin/sar (sar) in auto mode
update-alternatives: skip creation of /usr/share/man/man1/sar.1.gz because associated file /usr/share/man/man1/sar.sysstat.1.gz (of link group sar) doesn't exist
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-collect.timer → /usr/lib/systemd/system/sysstat-collect.timer.
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-summary.timer → /usr/lib/systemd/system/sysstat-summary.timer.
Created symlink /etc/systemd/system/multi-user.target.wants/sysstat.service → /usr/lib/systemd/system/sysstat.service.
Processing triggers for man-db (2.12.0-4build2) ...
shreedhanam2019@cloudshell:~$ lscpu
Linux 6.6.137+ (cs-1094043149067-default)      05/15/2026      _x86_64_          (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.2%    0.3%   0.0%    0.0%    0.0%   99.5%
```

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal
Cloud Shell Editor
cloudshell x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.78    0.00   0.38   0.07    0.03   98.73

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0              0.95         51.93         0.00         0.00     435956         0         0
sda               4.80         196.98        30.80         0.00     1683797     258583         0
sdb               0.12         15.93         0.26         0.00      133782     2204         0

shreedhanam2019@cloudshell:~$ vmstat
procs -----memory-----swap-----io-----system-----cpu-----
r  b   swpd   free   buff   cache   si   so    bi   bo    in   cs   us   sy   id   wa   st   gu
0    0   0 12351484  88364 1896772  0    0    212   31   813   3   1   0  99   0   0   0
shreedhanam2019@cloudshell:~$ sar
Cannot open /var/log/sarstat/sar1: No such file or directory
Please check if data collecting is enabled
shreedhanam2019@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:           16382844      2333972     12349468          1140       1985200       14048872
Swap:              0              0              0
shreedhanam2019@cloudshell:~$ mpstat
Linux 6.6.137+ (cs-1094043149067-default)      05/15/2026      _x86_64_      (4 CPU)

01:00:34 AM  CPU    %user   %nice    %sys %iowait    %irq   %soft  %steal  %guest   %gnice   %idle
01:00:34 AM    0     0.78    0.00   0.35   0.07   0.00   0.03   0.03   0.00   0.00   98.73
shreedhanam2019@cloudshell:~$ uptime
01:00:44 up 2:20, 0 user, load average: 0.08, 0.06, 0.06
shreedhanam2019@cloudshell:~$ sudo apt install dstat -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'pcp' instead of 'dstat'
The following additional packages will be installed:
  gawk libpcp-archive1t64 libpcp-gui2t64 libpcp-import1t64 libpcp-mmvt64 libpcp-pmda-perl
  libpcp-pmda3t64 libpcp-trace2t64 libpcp-web1t64 libpcp3t64 libpfm4 libsigsegv2 pcp pcp-conf
  python3-pcp
Suggested packages:
  gawk-doc pcp-gui libpcp-import-perl bpftrace python3-bpfcc
The following NEW packages will be installed:
  gawk libpcp-archive1t64 libpcp-gui2t64 libpcp-import1t64 libpcp-mmvt64 libpcp-pmda-perl
  libpcp-pmda3t64 libpcp-trace2t64 libpcp-web1t64 libpcp3t64 libpfm4 libsigsegv2 pcp pcp-conf
```

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal
Cloud Shell Editor
cloudshell x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

0 upgraded, 15 newly installed, 0 to remove and 2 not upgraded.
Need to get 4,619 kB of archives.
After this operation, 22.3 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libsigsegv2 amd64 2.14-1ubuntu2 [15.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 gawk amd64 1:5.2.1-2build3 [463 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 pcp-conf amd64 6.2.0-1.1build4 [20.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp3t64 amd64 6.2.0-1.1build4 [313 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-pmda3t64 amd64 6.2.0-1.1build4 [45.4 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-pmda-perl amd64 6.2.0-1.1build4 [41.4 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-archive1t64 amd64 6.2.0-1.1build4 [10.6 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-gui2t64 amd64 6.2.0-1.1build4 [12.4 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-import1t64 amd64 6.2.0-1.1build4 [17.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-mmvt64 amd64 6.2.0-1.1build4 [13.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-trace2t64 amd64 6.2.0-1.1build4 [16.9 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcp-web1t64 amd64 6.2.0-1.1build4 [213 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpfm4 amd64 4.13.0+git32-g0d4ed0e-1 [414 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-pcp amd64 6.2.0-1.1build4 [76.5 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/universe amd64 pcp amd64 6.2.0-1.1build4 [2,946 kB]
Fetched 4,619 kB in 0s (22.1 MB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libsigsegv2:amd64.
(Reading database ... 121252 files and directories currently installed.)
Preparing to unpack .../libsigsegv2_2.14-1ubuntu2_amd64.deb ...
Unpacking libsigsegv2:amd64 (2.14-1ubuntu2) ...
Setting up libsigsegv2:amd64 (2.14-1ubuntu2) ...
Selecting previously unselected package gawk.
(Reading database ... 121259 files and directories currently installed.)
Preparing to unpack .../00-gawk_1%3a5.2.1-2build3_amd64.deb ...
Unpacking gawk (1:5.2.1-2build3) ...
Selecting previously unselected package pcp-conf.
Preparing to unpack .../01-pcp-conf_6.2.0-1.1build4_amd64.deb ...
Unpacking pcp-conf (6.2.0-1.1build4) ...
Selecting previously unselected package libpcp3t64.
Preparing to unpack .../02-libpcp3t64_6.2.0-1.1build4_amd64.deb ...
Unpacking libpcp3t64 (6.2.0-1.1build4) ...
Selecting previously unselected package libpcp-pmda3t64.
Preparing to unpack .../03-libpcp-pmda3t64_6.2.0-1.1build4_amd64.deb ...
Unpacking libpcp-pmda3t64 (6.2.0-1.1build4) ...
Selecting previously unselected package libpcp-pmda-perl.
Preparing to unpack .../04-libpcp-pmda-perl_6.2.0-1.1build4_amd64.deb ...
Unpacking libpcp-pmda-perl (6.2.0-1.1build4) ...
```



```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

shreedhanam2019@cloudshell:~$ dstat
You did not select any stats, using -cdngy by default.
Error Parsing ASCII PMNS: Cannot open "/var/lib/pcp/pmns/root"
Failed to register derived metric:
PMNS not accessible <pcp.pmapl.c_uint_Array_1 object at 0x78213938cd50>.

shreedhanam2019@cloudshell:~$ sudo apt install glances -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  blt fonts-lyx isympy-common isympy3 javascript-common libblas3 libdouble-conversion3
  libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0 libopenjp2-7 libqhull-r8.0 libraqm0
  libtk8.6 libtk8.6-libs libwebpdemux2 libwebpmux3 libxft2 libxss1 lm-sensors
  python-matplotlib-data python3-appdirs python3-attr python3-bottle python3-brotli python3-bs4
  python3-certifi python3-chardet python3-contourpy python3-cssselect python3-cycler
  python3-dateutil python3-decorator python3-defusedxml python3-docker python3-fonttools
  python3-fs python3-html5lib python3-idna python3-influxdb python3-kiwisolver python3-lxml
  python3-lz4 python3-matplotlib python3-mpmath python3-msgpack python3-netifaces python3-numpy
  python3-olefile python3-pil python3-pil.imagetk python3-ply python3-psutil python3-pyasnl
  python3-pycryptodome python3-pysmi python3-pysnmp4 python3-pystache python3-requests
  python3-scipy python3-soupsieve python3-sympy python3-tk python3-tz python3-ufolib2
  python3-urllib3 python3-unicodedata2 python3-urllib3 python3-webencodings python3-websocket
  tk8.6-blt2.5 unicode-data
Suggested packages:
  blt-demo glances-doc python3-pynvim apache2 | lighttpd | httpd libjs-jquery-ui-docs tcl8.6
  tk8.6-fancontrol read-edid i2c-tools python-attr-doc python-cycler-doc python3-genshi
  python-lxml-doc cm-super-minimal dvipng ffmpeg fonts-staypuft ghostscript gir1.2-gtk-3.0
  inkscape ipython3 librsvg2-common python3-cairocffi python3-gi-cairo python3-gobject
  python3-pyqt5 python3-sip python3-tornado texlive-extra-utils texlive-latex-extra
  python-mpmath-doc python3-gmpy2 gfortran python3-pytest python-pil-doc python-ply-doc
  python3-openssl python3-socks python-requests-doc python-scipy-doc texlive-fonts-extra
  python-sympy-doc tix python3-tk-dbg
The following NEW packages will be installed:
  blt fonts-lyx glances isympy-common isympy3 javascript-common libblas3 libdouble-conversion3
  libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0 libopenjp2-7 libqhull-r8.0 libraqm0
  libtk8.6 libtk8.6-libs libwebpdemux2 libwebpmux3 libxft2 libxss1 lm-sensors
  python-matplotlib-data python3-appdirs python3-attr python3-bottle python3-brotli python3-bs4
  python3-certifi python3-chardet python3-contourpy python3-cssselect python3-cycler
  python3-dateutil python3-decorator python3-defusedxml python3-docker python3-fonttools
  python3-fs python3-html5lib python3-idna python3-influxdb python3-kiwisolver python3-lxml
  python3-lz4 python3-matplotlib python3-mpmath python3-msgpack python3-netifaces python3-numpy
  python3-olefile python3-pil python3-pil.imagetk python3-ply python3-psutil python3-pyasnl
  python3-pycryptodome python3-pysmi python3-pysnmp4 python3-pystache python3-requests
  python3-scipy python3-soupsieve python3-sympy python3-tk python3-tz python3-ufolib2
  python3-urllib3 python3-unicodedata2 python3-urllib3 python3-webencodings python3-websocket
  tk8.6-blt2.5 unicode-data
```

```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

""Sends SNMP notification.
Setting up glances (3.4.0.3dfsg-1) ...
Creating system user 'glances'.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/glances.service → /usr/lib/systemd/system/glances.service.
Setting up libtk8.6:amd64 (8.6.14-1build1) ...
Setting up python3-bs4 (4.12.3-1) ...
Setting up python3-fs (2.4.16-3) ...
Setting up python3-pil:amd64 (10.2.0-1ubuntu1) ...
Setting up isympy3 (1.12-7) ...
Setting up python3-requests (2.31.0+dfsg-1ubuntu1) ...
Setting up python3-numpy (1:1.26.4+ds-6ubuntu1) ...
Setting up python3-contourpy (1.0.7-2build1) ...
Setting up liblbfgsb0:amd64 (3.0+dfsg.4-1build1) ...
Setting up tk8.6-blt2.5 (2.5.3+dfsg-7build1) ...
Setting up python3-scipy (1.11.4-6build1) ...
Setting up python3-influxdb (5.3.1-6) ...
/usr/lib/python3/dist-packages/influxdb/tests/client_test.py:527: SyntaxWarning: invalid escape sequence '\('
  "\n", 'u', 'ms', 's', 'm' or 'h')"
/usr/lib/python3/dist-packages/influxdb/tests/influxdb08/client_test.py:309: SyntaxWarning: invalid escape sequence '\('
  "Invalid time precision is given. \n", 'm', 'ms' or 'u')"
/usr/lib/python3/dist-packages/influxdb/tests/influxdb08/client_test.py:453: SyntaxWarning: invalid escape sequence '\('
  "Invalid time precision is given. \n", 's', 'm', 'ms' or 'u')"
/usr/lib/python3/dist-packages/influxdb/tests/influxdb08/client_test.py:737: SyntaxWarning: invalid escape sequence '\('
  "permissions" must be \n(readFrom, writeTo) tuple"
Setting up blt (2.5.3+dfsg-7build1) ...
Setting up python3-docker (5.0.3-1ubuntu1) ...
Setting up python3-tk:amd64 (3.12.3-0ubuntu1) ...
Setting up python3-pil.imagetk:amd64 (10.2.0-1ubuntu1) ...
Setting up python3-fonttools (4.46.0-1build2) ...
Setting up python3-ufolib2 (0.16.0+dfsg1-1) ...
Setting up python3-matplotlib (3.6.3-1ubuntu5) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for fontconfig (2.15.0-1ubuntu2) ...
Errors were encountered while processing:
 pc
E: Sub-process /usr/bin/dpkg returned an error code (1)
shreedhanam2019@cloudshell:~$ glances
shreedhanam2019@cloudshell:~$
```

CHAPTER 9

USER MANAGEMENT

```
shreedhanam2019@cloudshell:~$ sudo useradd testuser
shreedhanam2019@cloudshell:~$ cat /etc/passwd | grep testuser
testuser:x:1001:1001::/home/testuser:/bin/sh
shreedhanam2019@cloudshell:~$ sudo userdel testuser
shreedhanam2019@cloudshell:~$ cat /etc/passwd | grep testuser
shreedhanam2019@cloudshell:~$ usermod --help
Usage: usermod [options] LOGIN
```

Options:

-a, --append	append the user to the supplemental GROUPS mentioned by the -G option without removing the user from other groups
-b, --badname	allow bad names
-c, --comment COMMENT	new value of the GECOS field
-d, --home HOME_DIR	new home directory for the user account
-e, --expiredate EXPIRE_DATE	set account expiration date to EXPIRE_DATE
-f, --inactive INACTIVE	set password inactive after expiration to INACTIVE
-g, --gid GROUP	force use GROUP as new primary group
-G, --groups GROUPS	new list of supplementary GROUPS
-h, --help	display this help message and exit
-l, --login NEW_LOGIN	new value of the login name
-L, --lock	lock the user account
-m, --move-home	move contents of the home directory to the new location (use only with -d)
-o, --non-unique	allow using duplicate (non-unique) UID
-p, --password PASSWORD	use encrypted password for the new password
-P, --prefix PREFIX_DIR	prefix directory where are located the /etc/* files
-r, --remove	remove the user from only the supplemental GROUPS mentioned by the -G option without removing the user from other groups
-R, --root CHROOT_DIR	directory to chroot into
-s, --shell SHELL	new login shell for the user account
-u, --uid UID	new UID for the user account
-U, --unlock	unlock the user account
-v, --add-subuids FIRST-LAST	add range of subordinate uids
-V, --del-subuids FIRST-LAST	remove range of subordinate uids
-w, --add-subgids FIRST-LAST	add range of subordinate gids
-W, --del-subgids FIRST-LAST	remove range of subordinate gids
-Z, --selinux-user SEUSER	new SELinux user mapping for the user account

```

shreedhanam2019@cloudshell:~$ passwd --help
Usage: passwd [options] [LOGIN]

Options:
  -a, --all                report password status on all accounts
  -d, --delete             delete the password for the named account
  -e, --expire             force expire the password for the named account
  -h, --help              display this help message and exit
  -k, --keep-tokens        change password only if expired
  -i, --inactive INACTIVE set password inactive after expiration
                           to INACTIVE
  -l, --lock              lock the password of the named account
  -n, --mindays MIN_DAYS  set minimum number of days before password
                           change to MIN_DAYS
  -q, --quiet             quiet mode
  -r, --repository REPOSITORY change password in REPOSITORY repository
  -R, --root CHROOT_DIR   directory to chroot into
  -S, --status            report password status on the named account
  -u, --unlock            unlock the password of the named account
  -w, --warndays WARN_DAYS set expiration warning days to WARN_DAYS
  -x, --maxdays MAX_DAYS set maximum number of days before password
                           change to MAX_DAYS

shreedhanam2019@cloudshell:~$ sudo chage -l $USER
Last password change                : May 15, 2026
Password expires                    : never
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
shreedhanam2019@cloudshell:~$ groups
shreedhanam2019 adm sudo docker
shreedhanam2019@cloudshell:~$ sudo groupadd testgroup
shreedhanam2019@cloudshell:~$ sudo groupdel testgroup
shreedhanam2019@cloudshell:~$ groups
shreedhanam2019 adm sudo docker
shreedhanam2019@cloudshell:~$ su --help

Usage:
  su [options] [-] [<user> [<argument>...]]

```

```

shreedhanam2019@cloudshell:~$ su --help

Usage:
  su [options] [-] [<user> [<argument>...]]

Change the effective user ID and group ID to that of <user>.
A mere - implies -l. If <user> is not given, root is assumed.

Options:
  -m, -p, --preserve-environment    do not reset environment variables
  -w, --whitelist-environment <list> don't reset specified variables

  -g, --group <group>               specify the primary group
  -G, --supp-group <group>          specify a supplemental group

  -, -l, --login                    make the shell a login shell
  -c, --command <command>           pass a single command to the shell with -c
  --session-command <command>      pass a single command to the shell with -c
                                   and do not create a new session
  -f, --fast                        pass -f to the shell (for csh or tcsh)
  -s, --shell <shell>               run <shell> if /etc/shells allows it
  -P, --pty                         create a new pseudo-terminal

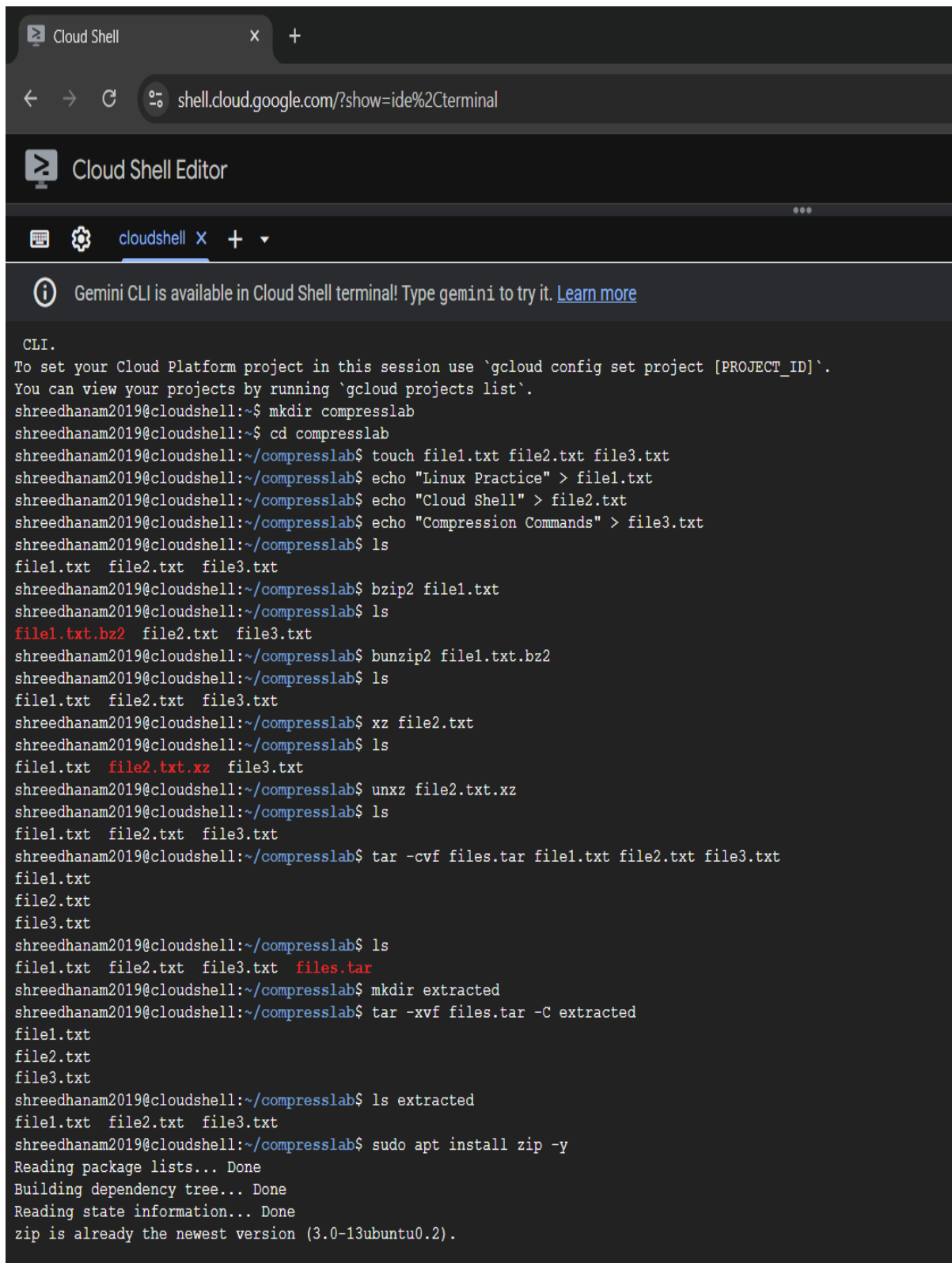
  -h, --help                        display this help
  -V, --version                     display version

For more details see su(1).
shreedhanam2019@cloudshell:~$ █

```

CHAPTER 10

COMPRESSION AND ARCHIVING



```
Cloud Shell
shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

cloudshell X +

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
shreedhanam2019@cloudshell:~$ mkdir compresslab
shreedhanam2019@cloudshell:~$ cd compresslab
shreedhanam2019@cloudshell:~/compresslab$ touch file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ echo "Linux Practice" > file1.txt
shreedhanam2019@cloudshell:~/compresslab$ echo "Cloud Shell" > file2.txt
shreedhanam2019@cloudshell:~/compresslab$ echo "Compression Commands" > file3.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ bzip2 file1.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt.bz2 file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ bunzip2 file1.txt.bz2
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ xz file2.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt file2.txt.xz file3.txt
shreedhanam2019@cloudshell:~/compresslab$ unxz file2.txt.xz
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ tar -cvf files.tar file1.txt file2.txt file3.txt
file1.txt
file2.txt
file3.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
file1.txt file2.txt file3.txt files.tar
shreedhanam2019@cloudshell:~/compresslab$ mkdir extracted
shreedhanam2019@cloudshell:~/compresslab$ tar -xvf files.tar -C extracted
file1.txt
file2.txt
file3.txt
shreedhanam2019@cloudshell:~/compresslab$ ls extracted
file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ sudo apt install zip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
zip is already the newest version (3.0-13ubuntu0.2).
```

 Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

```
zip is already the newest version (3.0-13ubuntu0.2).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Setting up pcp (6.2.0-1.1build4) ...
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
dpkg: error processing package pcp (--configure):
 installed pcp package post-installation script subprocess returned error exit status 1
Errors were encountered while processing:
 pcp
E: Sub-process /usr/bin/dpkg returned an error code (1)
shreedhanam2019@cloudshell:~/compresslab$ zip myfiles.zip file1.txt file2.txt file3.txt
  adding: file1.txt (stored 0%)
  adding: file2.txt (stored 0%)
  adding: file3.txt (stored 0%)
shreedhanam2019@cloudshell:~/compresslab$ ls
extracted file1.txt file2.txt file3.txt files.tar myfiles.zip
shreedhanam2019@cloudshell:~/compresslab$ mkdir unzipfolder
shreedhanam2019@cloudshell:~/compresslab$ unzip myfiles.zip -d unzipfolder
Archive: myfiles.zip
  extracting: unzipfolder/file1.txt
  extracting: unzipfolder/file2.txt
  extracting: unzipfolder/file3.txt
shreedhanam2019@cloudshell:~/compresslab$ ls unzipfolder
file1.txt file2.txt file3.txt
shreedhanam2019@cloudshell:~/compresslab$ lzma file3.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
extracted file1.txt file2.txt file3.txt.lzma files.tar myfiles.zip unzipfolder
shreedhanam2019@cloudshell:~/compresslab$ sudo apt install lzop -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  liblz2-2
The following NEW packages will be installed:
  liblz2-2 lzop
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
1 not fully installed or removed.
Need to get 136 kB of archives.
After this operation, 322 kB of additional disk space will be used.
```

```
Cloud Shell
x +
← → ↺ shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor
cloudshell x + ▾

(i) Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

liblz2-2
The following NEW packages will be installed:
liblz2-2 lzop
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
1 not fully installed or removed.
Need to get 136 kB of archives.
After this operation, 322 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 liblz2-2 amd64 2.10-2build4 [54.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 lzop amd64 1.04-2build3 [82.2 kB]
Fetched 136 kB in 0s (2,609 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package liblz2-2:amd64.
(Reading database ... 132380 files and directories currently installed.)
Preparing to unpack .../liblz2-2_2.10-2build4_amd64.deb ...
Unpacking liblz2-2:amd64 (2.10-2build4) ...
Selecting previously unselected package lzop.
Preparing to unpack .../lzop_1.04-2build3_amd64.deb ...
Unpacking lzop (1.04-2build3) ...
Setting up pcsp (6.2.0-1.1build4) ...
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
dpkg: error processing package pcsp (--configure):
 installed pcsp package post-installation script subprocess returned error exit status 1
Setting up liblz2-2:amd64 (2.10-2build4) ...
Setting up lzop (1.04-2build3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Errors were encountered while processing:
 pcsp
E: Sub-process /usr/bin/dpkg returned an error code (1)
shreedhanam2019@cloudshell:~/compresslab$ touch fast.txt
shreedhanam2019@cloudshell:~/compresslab$ lzop fast.txt
shreedhanam2019@cloudshell:~/compresslab$ ls
extracted fast.txt.lzo file2.txt files.tar unzipfolder
fast.txt file1.txt file3.txt.lzma myfiles.zip
shreedhanam2019@cloudshell:~/compresslab$ unlzop fast.txt.lzo
-bash: unlzop: command not found
shreedhanam2019@cloudshell:~/compresslab$ ls
extracted fast.txt.lzo file2.txt files.tar unzipfolder
fast.txt file1.txt file3.txt.lzma myfiles.zip
shreedhanam2019@cloudshell:~/compresslab$
```